

Blockchain-Based Key Management and Authentication Scheme for IoT Networks With Chaotic Scrambling

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Abstract—Internet of Things (IoT) systems have the great advantage of connecting massive low energy-consumption devices for the information delivery via wireless channels. However, IoT devices suffer from malicious attacks or eavesdropping due to the broadcasting property of wireless channels. With the aim of enhancing the security performances, we first propose a chaotic scrambler, which utilizes the natural high security property of chaotic sequences being aperiodic and sensitive to initial values, to improve the security of signal transmissions. Then, we propose to utilize the traceable and tamper-proof blockchain to integrate the confidential chaotic key into the chain for user identity management and authentications. Subsequently, theoretical security, reliability and throughput capacities are analyzed, and simulation results are provided to validate the design and the theoretical analysis. Furthermore, we present the practical hardware implementation board for the proposed secure chaotic scrambling following the long range wide area network (LoRaWAN) specifications, which demonstrates that the security performances of IoT devices can be effectively enhanced.

Index Terms—Chaos-based scrambler, hash function and blockchain, Internet of Things, key management and authentication mechanism, security.

I. INTRODUCTION

INTERNET of Things (IoT) has great advantages of bridging the cyberspace of digital systems and heterogeneous physical terminals to achieve large scale machine-to-machine (M2M) communications, which is also categorized as machine-type communications (MTC) in the Third Generation

Partnership Project Committee (3GPP) as one of the most potential scenarios of the Fifth Generation (5 G) networks [1]. Along with the exponentially increasing demands for the information delivery between diverse user equipments (UEs), IoT devices have been widely deployed to construct low power wide area network (LPWAN) to provide ubiquitous services and applications, such as industrial monitoring, health care, automatic vehicles etc. [2].

However, due to the natural broadcasting property of wireless channels, hackers could easily initiate malicious attacks to disrupt wireless information transmissions, or implement eavesdropping to steal the user data. Besides, IoT system is vulnerable to camouflage and impersonating attack since each heterogeneous UE can access the IoT network due to the broadcasting property of wireless channels. IoT attacks can be classified as passive and active attacks based on whether the transmissions are interrupted or not [3], while attackers can launch an attack at different layers such as the physical layer or the application layer [4]. For example, near field attacks, including jamming attack, replay attack, tampering attack and distributed denial of service (DDoS) attack [4], [5], are launched at the physical layer within a short range, which might lead to severe security performance degradations.

The narrow-band Internet of Things (NB-IoT) and the long range wide area network (LoRaWAN) are two main competitive protocols to provide secure information transmissions for massive UEs via wireless channels, having met the requirements of long range, low energy consumptions [6]. Although the two types of IoT systems have provided security enhancement protocols at higher layers, they still need to combat different types of attacks. For example, LoRaWAN might undergo security threats [7], such as the DDoS attack [8], the replay attack [9] or the jamming attack [10].

Therefore, to combat the information leakage and identity camouflage, a lot of promising research works have been done to enhance the security performances at different layers for IoT systems. For example, the blockchain-based encryption scheme has been applied at the application layer [11] to provide tamper-proof and traceable transmissions for secure IoT communications. Besides, at the network layer, the deep learning scheme has been presented to combat various attacks intelligently [12].

Nevertheless, all the security mechanisms design at upper layers than the physical layer can not mitigate the attacks at

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the physical layer when IoT devices undergo the near-field attacks, since attackers can easily disrupt the information transmissions via broadcasting wireless channels. Accordingly, at the physical layer, how to enhance the security performances of IoT devices, which have constrained battery life and storage space as well as limited computational capabilities, has become a challenging issue to be investigated.

A. Related Works

To enhance the physical layer security performances, in recent research works, the authors in [13] proposed the signal processing remedies to cope with the security threats, while the researchers in [14] proposed a cooperative jammer to fight against multiple passive and non-colluding eavesdroppers with the artificial noise aided secrecy beamforming. In our previous work [15], we proposed a radio frequency (RF) watermark method to embed the secret information into constellation points. Besides, the continuous dynamical chaotic map, such as Chebyshev polynomial function, have been proposed for secure transmission [16], [17]. Then, the iterative discrete-time representation of dynamical systems for chaos generators are designed for cryptographic applications [18], [19]. However, the chaos synchronization is complex and can be hardly implemented in wireless communications, and the confidential initial conditions are delivered in a specific channel, which might be intercepted over broadcasting wireless channels.

Except for the signal processing solutions, the identity authentication mechanism can be further applied to enhance the security performance. Considering that the intrinsic physical characteristics of components are unique for each device, the researchers in [20] proposed to formulate the physical unclonable function (PuF) to generate the unique hardware identity for IoT devices. Moreover, to authenticate the identity, the blockchain has been applied in IoT networks to provide a single source of trust and to improve security, reliability, auditability and scalability performances. Blockchain is a distributed computing, decentralized and peer-to-peer (P2P) data sharing technology, and could provide the cryptography and consensus mechanism to achieve secure, traceable and interoperable information delivery.

Recently, in [21], the authors have proposed a physical layer authentication method based on the soft information, wherein the authentic key is randomized based on the hash mapping, while in [22][23], the traceable and tamper-proof property of the blockchain is exploited to improve the security performances. Additionally, in [24], the authors propose a blockchain-aided identity management system, while in [25], an out-of-band two factor authentication scheme has been proposed to store the device information via the blockchain. Besides, Lin et al. [26] propose a blockchain-based solution to guarantee the trust and validate the data transmission in Join Procedure for LoRaWAN networks. Then based on this research achievement, Danish et al. [27] propose a blockchain-aided two factor authentication mechanisms to further enhance the security performances.

However, most of existing works utilize linear such as the watermarking to enhance the security, while the repetitive linear operations might be learned by attackers for launching attacks. Besides, few research works have been conducted on the practical implementation of the security enhancement scheme.

B. Our Work and Contributions

Our motivation is to improve the physical layer security performances by utilizing the natural high security properties of chaotic sequences being aperiodic and sensitive to the initial value, and the tamper-proof property of hash functions to deliver the secret chaotic key. In this paper, we propose a blockchain-enabled chaos-based scrambling IoT (BCS-IoT) scheme to enhance the security of transmissions.

Different from the linear security enhancement schemes such as the watermarking or the artificial noise imposing, in our design, we first utilize the non-linear chaotic signal processing to hide the user data behind the irregular chaotic behavior, then we propose a blockchain-based mechanism to manage and authenticate the secret chaotic key for legitimate receivers to guarantee the secure and traceable information delivery in heterogeneous IoT communications.

To be more explicit, we first generate a chaotic sequence with the secret initial value and mapping pattern. Then we construct the frequency hopping matrix based on this chaotic sequence to scramble the transmitted signals. As a result, the information-bearing signals are transmitted over wireless channels hopping in a chaotic way. Notably, the chaotic behavior is determined by the initial value and chaotic mapping pattern, hence they are regarded as the secret chaotic keys.

Then we propose to utilize the tamper-proof blockchain to enable the management and authentication of secret chaotic keys. At the receiver side, only the legitimate users can retrieve the received data with the extracted known initial value of the chaotic sequence. By contrast, eavesdroppers or malicious users can hardly obtain the transmitted data, since they can not get authenticated and no secret initial value embedded in the blockchain is available for the received signal retrieval. Accordingly, by exploiting the blockchain enabled chaotic scrambling, the security performances can be enhanced to combat malicious attacks.

We provide theoretical reliability, security and throughput analysis. The simulation results are presented to analyze the framing complexity, delay time, system throughput, reliability and security performances. Moreover, we provide the hardware implementation board and compose a practical blockchain-enabled chaotic scrambler to validate the proposed design.

Briefly, the main contributions are presented as follows.

- 1) In order to combat near-field attacks and enhance physical security performances, we propose to utilize and construct an aperiodic chaos-based scrambling matrix to hide user data behind irregular chaotic waveforms.
- 2) We propose to construct the blockchain to manage the secret key, i.e., the initial value, which is embedded into the block-hash by using the secure Hash algorithm

(SHA). Then the legitimate receivers can extract the key after the authentication, while eavesdroppers or malicious users can not retrieve the data without the consensus. Hence the security performances can be improved.

- 3) We propose an authentication mechanism to verify the identity of users based on the blockchain. Then we derive theoretical authentication error rate (AER), the mutual information and secrecy capacity expressions of legitimate users and two-type eavesdroppers.
- 4) We construct the practical hardware implementations of the proposed design to validate our design.

C. Organization and Notations

The rest of this paper is organized as follows. In Section II, we present the details of the proposed BCS-IoT system. Then in Section II-D and Section II-B, we introduce the blockchain aided authentication and the chaos-based scrambling mechanism in detail. Subsequently, Section III provides theoretical performance analysis and Section IV presents simulation results as well as the implementation of practical systems. Finally Section V concludes our findings.

For brevity of expressions, we use lowercase letters in boldface to denote the vector and uppercase letters in boldface for matrix. Besides, $\lceil \cdot \rceil$ denotes the round up operation, and $\mathbb{B}, \mathbb{R}, \mathbb{C}$ respectively denote the set of binary, real and complex number. In addition, $(\hat{\cdot})$ denotes the estimation at the receiver, $\star$ denotes the convolution operation, $\| \cdot \|_2$ denotes the 2-norm of a vector, $(\cdot)^T$ denotes the transposition and $\mathbb{E}(\cdot)$ denotes the numerical expectation of a random variable. The function $Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = \frac{1}{2} \text{erfc}(\frac{\sqrt{2}}{2}x)$ denotes the right tail function of the standard normal distribution.

II. BLOCKCHAIN-ENABLED CHAOTIC SCRAMBLING IOT SYSTEM DESIGN

In this section, we present the details of the blockchain-enabled chaotic scrambling IoT system design. Without loss of generality, we mainly focus on the LoRaWAN IoT network, which operates over the unlicensed Industrial Scientific Medical (ISM) spectrum bands with a star or net topology. With the broad bandwidth provided by the chirp spread spectrum (CSS) modulation scheme, the LoRaWAN transmits confidential message through a continuously varying carrier frequency, which supports the data rate from 0.3kbps to 50kbps and 2-20 km transmission coverage with low power cost [28].

In our design, the user data are transmitted at the frequency which hops according to a chaos-based scrambling matrix, while the confidential information is embedded into a blockchain for authentications. At the stage of the proposed Join Procedure, we generate the digital signature of the secure chaotic keys by using the SHA. The digital signature will then be registered in a new block and activates the random-searching process (also called mining in the following paragraph) to implant the secret chaotic keys into ciphertext as the block-hash. Then given the structure of blockchain, the new block is linked successively as a Hash Chain. After being transmitted via wireless channels,

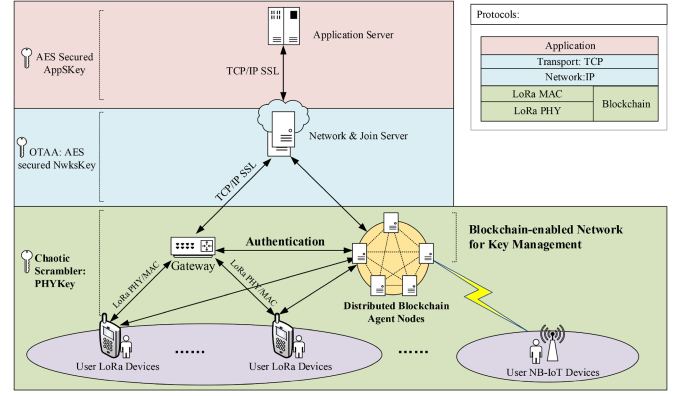


Fig. 1. Architecture of blockchain-enabled chaotic scrambling IoT (LoRaWAN) system.

the legitimate receiver will get authenticated using the device information in the blockchain with the consensus mechanism. Thus the secret chaotic key will be extracted for further information retrieval. Next, at the stage of the Transmit Procedure, the information-bearing signals based on chaotic scrambling matrix are transmitted at the aperiodically hopping frequency.

It is worth pointing out that the following proposed design could also be extended to the NB-IoT network, in which the authentication in the Join Procedure is conducted in Control Plane and the communication in the Transmit Procedure is done by User Plane.

A. System Architecture

The Fig. 1 presents the architecture of our proposed blockchain-enabled chaotic scrambling IoT system, wherein we establish an extra blockchain layer and utilize the authentication mechanism to manage the secret initial value. Afterwards, the data will be transmitted at the frequency hopping in a chaotic way. At the receiver side, the identify of IoT devices is authenticated by using the dual authentication mechanism [27]. Thus only the legitimate users can retrieve the delivered secret key for further signal processing. By contrast, the malicious users can not obtain the transmitted information without the secret knowledge.

Notably, different from the traditional LoRaWAN system, in our considered network, the device information is delivered to both the server and to the agent nodes which act as the transceivers between LoRaWAN and blockchain network. This identity information of devices will be used to activate the registration, and to complete the authentication between gateway and server via the confidential chaotic key.

To be more explicit, in the proposed BCS-LoRaWAN system, the blockchain acts as a distributed computing database and decentralized point-to-point (P2P) network, which does not involve any implementation of trusted third-party servers [22]. Thanks to the application of the hash chain structure, cryptography, consensus mechanism and distributed networking, the blockchain could enable the LoRaWAN system to be decentralized, tamper-proof, traceable and scalable for heterogeneous accessing.

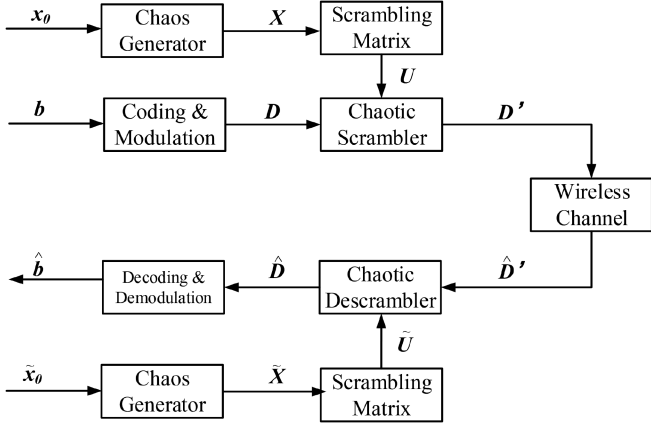


Fig. 2. Diagram of the blockchain-enabled chaotic scrambling scheme.

Next, we will introduce how to integrate the secret chaotic key into blockchain. Then we will present how to implement the authentication in the Join Procedure, and how to scramble the user data in a chaotic way in the Transmit Procedure.

B. Transmit Procedure

The Fig. 2 illustrates the transceiver model of the proposed blockchain-enabled chaotic scrambling system in the Transmit Procedure. At the transmitter, the chaos generator outputs the chaotic sequence according to a specific mapping pattern. Here we use the second order Chebyshev polynomial function (CPF) to generate the chaotic sequences $\mathbf{x} = [x_0, \dots, x_\beta]$ of the length β by

$$x_{i+1} = 1 - 2x_i^2, \quad 0 \leq k \leq \beta - 1 \quad (1)$$

where x_i denotes the i^{th} chip, and x_0 represents the input initial value which is the secret chaotic key managed in the Join Procedure above. Then we construct the chaotic matrix $\mathbf{X} = [\mathbf{x}_0, \dots, \mathbf{x}_{N_{slot}-1}]$ with the multiple chips $\mathbf{x}_n = [x_{0,n}, \dots, x_{N_{sc}-1,n}]^T$.

On the other hand, the user data bits $\mathbf{b} \in \mathbb{B}$ are coded and modulated into a data frame $\mathbf{D} \in \mathbb{C}^{N_{sc} \times N_{slot}}$ which uses the number of subcarriers N_{sc} and the number of time slots N_{slot} . The data frame $\mathbf{D}$ contains multiple symbols, which can be expressed by $\mathbf{D} = [\mathbf{D}_0, \dots, \mathbf{D}_{N_{slot}-1}]$, while the symbol at the k^{th} slot is denoted by $\mathbf{D}_k = [d_{0,k}, \dots, d_{N_{sc}-1,k}]^T$.

Subsequently, we construct the scrambling matrix $\mathbf{U}_k$ for scrambling the k^{th} slot symbols $\mathbf{D}_k$, which is determined by the k^{th} chaotic sequence $\mathbf{x}_k$. Thus, the user data will be scrambled as $\mathbf{D}' = [\mathbf{D}'_0, \dots, \mathbf{D}'_{N_{slot}-1}]^T$, which is composed with $\mathbf{D}'_k = [d'_{0,k}, \dots, d'_{N_{sc}-1,k}]$ via [29]:

$$\mathbf{D}'_k = \mathbf{U}_k \mathbf{D}_k; \mathbf{D}_k \in \mathbb{R}^{N_{sc} \times 1}, \mathbf{U}_k \in \mathbb{B}^{N_{sc} \times N_{sc}} \quad (2)$$

Notably, for $\mathbf{U}_k$, there is only one “1” in each row and column, while the position of “1” varies in a chaotic way. We then apply the scrambling mechanism presented in our previous work [30] to derive $\mathbf{U}_k$ by (3).

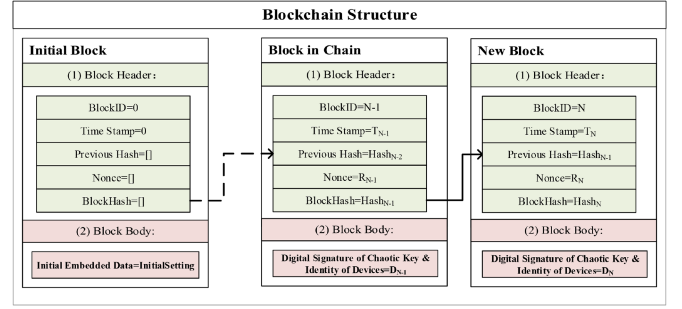


Fig. 3. Diagram of blockchain with the confidential chaotic key embedded.

$$\begin{aligned} \mathbf{U}_k &= f_S^{sort}(\mathbf{U}_k^{sort}) = f_S^{sort}([\mathbf{x}_k]^{-1} \mathbf{x}_k^{sort}) \\ &\rightarrow \{\mathbf{U}_k\}_{i,j} = \begin{cases} 1 & , \{\mathbf{U}_k^{sort}\}_{i,j} = 1 \\ 0 & , \text{elsewise} \end{cases} \end{aligned} \quad (3)$$

where $[\mathbf{x}_k]^{-1} = [\frac{1}{x_{0,k}}, \dots, \frac{1}{x_{N_{sc}-1,k}}]^T$, and $\mathbf{x}_k^{sort} = [x_1^{sort}, \dots, x_{N_{sc}}^{sort}]$ is the sorted version of $\mathbf{x}_k$ which meets $\forall i < j, x_i^{sort} \leq x_j^{sort}$. Besides, $f_S^{sort}(\cdot)$ is a function that represents the index after sorting the chaotic sequence.

When $\mathbf{X}$ keeps invariant in several slots, namely $\forall i, j: \mathbf{x}_i = \mathbf{x}_j = \mathbf{x}$, the scrambling matrix becomes a unified frequency hopping(FH) matrix $\mathbf{U}$ given as below.

$$\mathbf{U} = f_S^{sort}([\mathbf{x}]^{-1} \mathbf{x}^{sort}), [\mathbf{x}]^{-1} = \left[\frac{1}{x_0}, \dots, \frac{1}{x_{N_{sc}-1}} \right]^T \quad (4)$$

Thereafter at the legitimate receiver, the chaotic scrambling matrix $\tilde{\mathbf{U}}_k$ can be recovered with the learned secret key, then the received signal $\mathbf{D}'$ can be obtained via the de-scrambling. After decoding and demodulation, the receiver can obtain the user data $\hat{\mathbf{b}}$.

C. Blockchain Construction

The Fig. 3 illustrates the data structure of a blockchain, wherein the secret key, i.e. the initial value of chaotic sequences, are embedded. We can notice that the blockchain consists of blocks sequentially connected with each other.

Each block in the blockchain is composed by two parts: body and header, while the confidential chaotic key and identity information are registered and stored in the body part. Considering that the storage space of blocks is limited, Merkle Tree algorithm [22] can be applied to divide confidential data into several pieces, which are then abstracted and assembled based on the tree structure.

On the other hand, for the header part, the time-stamp and blockID, which index a block in this hash-chain, are fundamental elements to guarantee the one-way chain assembled in the time order and the irreversibility of blockchain. Besides, since the previous block-hash is also inherited in a new block, any bit-tamper on the previous blocks will affect the whole chain, thus integrity, immutability and provenance of confidential data could be guaranteed.

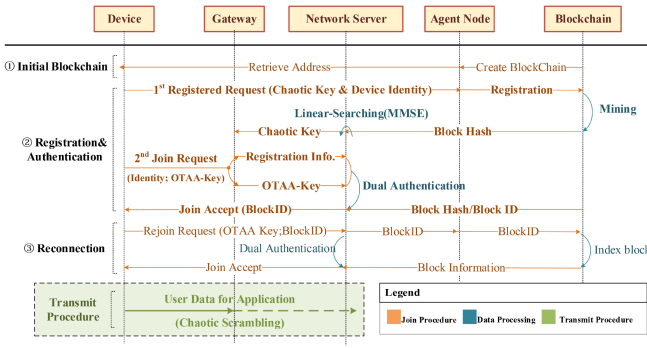


Fig. 4. Flowchart of blockchain-enable management and dual authentication scheme for LoRaWAN.

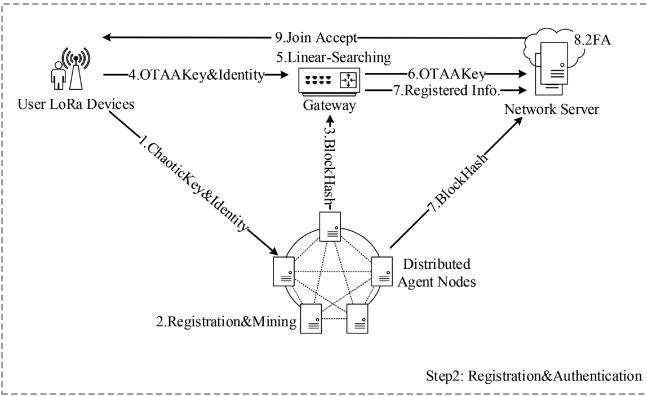


Fig. 5. Step 2: Registration & authentication in the join procedure.

Moreover, we design a new block generation condition as a mathematical puzzle to embed the confidential chaotic key into the block-hash by conducting a random search (also called mining) among distributed nodes. In other words, we propose a mining condition to enable the distributed nodes to generate a nonce to implement the block-hash with the first label and the index-flipping by the chaotic key. Finally, in order to synchronize in a distributed heterogeneous networking, we utilize the proof of work (PoW) consensus mechanism [31]. The first distributed edge-node which accomplishes the mining process, links the new block and broadcasts the new chain to the network.

D. Join Procedure

Next, with proposed blockchain-enable management and authentication design, a new device could access a IoT network (e.g. LoRaWAN) after the Join Procedure as below.

The Fig. 4 illustrates the dual authentication procedure with the aid of the blockchain. In the Join Procedure, the new LoRa device will be authenticated and get access to the network. We divide the accessing process into three steps to be detailed as below, which are step1: initialization, step2: registration and authentication and step3: reconnection.

1) *Three Steps for Access to IoT Networks: Step 1: Initialization.* In this step, distributed agent nodes activate and generate the initial block to initialize the blockchain-

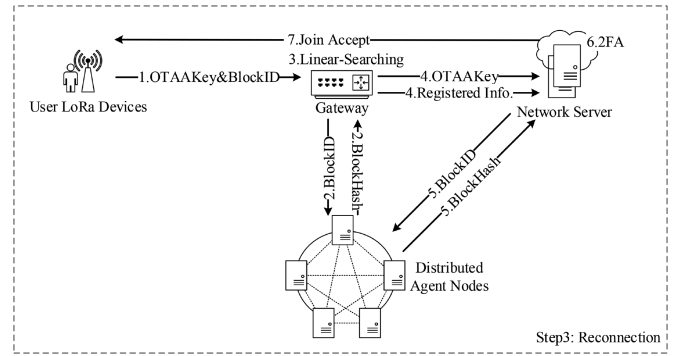


Fig. 6. Step 3: Reconnection in the join procedure.

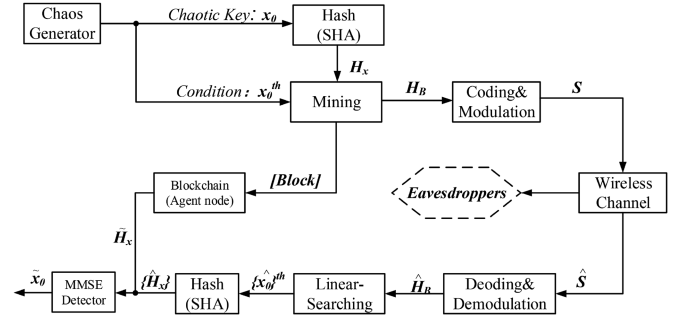


Fig. 7. The transceiver model for confidential information transmissions of blockchain-enable authentication scheme in the Join Procedure.

enable authentication network. Then the address of running this network is sent back to the gateway and network server to allow accessing the registered information in the blockchain, and to the devices for the information reception.

Step 2: Registration & Authentication. Fig. 5 illustrates that when a new device needs to access the network, a registration request frame is composed and delivered to distributed agent nodes in blockchain for registration and key-embedding. Meanwhile, a join request frame is composed and delivered to the network server via the gateway for LoRaWAN over-the-air activation (OTAA) authentication. After generating the new block, the block-hash is delivered to the gateway and the network server. Gateways conduct linear-searching process with the minimum mean square error (MMSE) module to obtain the embedded confidential chaotic key in the received block-hash. Then the OTAA and registration frame composed by gateways is sent to the network server. After authentication, the block ID will be sent back to devices in join accept frame.

Step 3: Reconnection. As shown in Fig. 6, after the device completes the authentication, the user data can be delivered in the Transmit Procedure. Then in order to reduce the signaling consumption, the device can reconnect the network via sending the request frame with the block-ID to the network server for the next authentication.

2) *Blockchain-Enabled Authentication and Confidential Information Transmissions:* The Fig. 7 presents the transceiver model used for the confidential information transmissions of proposed blockchain-enabled authentication scheme.

Algorithm 1: Generate new blocks.

```

1: Initialize Flag:  $cond = false$ ;
2: Initialize Nonce:  $R = MineStart$ ;
3: Encrypted Chaotic Key:  $H_x = H^S(x_0)$ ;
4: Set Mining Condition:  $mc(H_B)$ 
   Consition1:  $\overline{H_B(1 : L_{Mine})} = MineStr?$ 
   Consition2:  $H_B(L_{Mine} + 1 + x_0) = 0?$ 
5: while  $cond = false$  do
6:   Linear step-up:  $R = R + MineStep$ ;
7:   Compose block:  $B = String(\{Header|_R, H_x\})$ ;
8:   Calculate block-hash:  $H_B = H^{bc}(B)$ 
9:   if  $mc(H_B) = True$  then
10:     $cond = true$ ;  $break$ ;
11:   end if
12: end while
13: Consensus mechanism(e.g.PoW):  $H_B^N = H_B$ 
    
```

At the transmitter, the secret chaotic key of legitimate user x_0 is firstly abstracted by SHA function $H^s(\cdot)$ to generate the digital signature $H_x = H^s(x_0) \in \mathbb{B}^{1 \times L}$ of the length L , which is stored in the block-body, and thus the registration can be realized. Thanks to the aperiodic property of chaotic sequences, the space can be large enough to support massive connections of heterogeneous IoT networks. Without loss of generality, to facilitate the practical implementations, we can set the secret chaotic key that consists of the mapping pattern and initial values as an integer $x_0 \in \mathbb{Z}$.

Thereafter, distributed agent nodes will be activated random-searching to calculate the block-hash $H_B \in \mathbb{B}^{1 \times L_B}$ of the length L_B to reach the designed predefined conditions, which can be expressed as (5) below.

1. The first L_{Mine} bits in the block-hash H_B are marked as $\overline{H_B(1 : L_{Mine})}$ which is predefined as the target **MineStr**.
2. The $(L_{Mine} + x_0 + 1)^{th}$ bits in H_B are flipped to 0, which can embed x_0 into H_B .

$$\begin{aligned}
 R : H_B &= H^{bc}(B|_R) = H^{bc}(\{Header|_R, H_x\}) \\
 s.t. \begin{cases} \overline{H_B(1 : L_{Mine})} &= \text{MineStr} \\ H_B(L_{Mine} + x_0 + 1) &= 0 \end{cases}
 \end{aligned} \quad (5)$$

where $Header|_R$ and $B|_R$ respectively represent the composed header and block data based on the structure shown in Fig. 3 with the random nonce R .

The generation procedure is detailed in Algorithm 1 as follows, where $String(\cdot)$ converts the data into a string array, $MineStart$ and $MineStep$ are parameters used for the mining process, and H_B^i is the i^{th} block. Then the physical chaotic keys are registered and embedded into the blocks linked as a hash chain as below

$$H_B^N \leftarrow H_B^{N-1} \leftarrow \dots \leftarrow H_B^1 \leftarrow H_B^0 \quad (6)$$

where H_B^0 is the initial block generated during the initialization procedure.

After the mining and integrating the confidential chaotic key, we apply the cyclic-shifting chirp spread spectrum (CS-CSS)

Algorithm 2: Retrieve the chaotic key at the legitimate receiver.

```

1: for  $i = 0 : L_B - L_{Mine} - 1$  do
2:   if  $\hat{H}_B(i) = 0$  then
3:      $\hat{H}_x = H^s(L_{Mine} + 1 + i)$ ;
4:      $\hat{x}_0 = \min_i(\|\hat{H}_x - H_x\|_2)$ ;
5:   end if
6: end for
    
```

module to transmit the user data as specified by the LoRaWAN standard. In every frame, the spectrum bands are separated into 2^{SF} chips, while the user data are modulated onto the chips, where SF denotes the spreading factor. Note that $SF \in [7, 8, 9, 10, 11, 12]$ as specified by the LoRaWAN standard. With considerations of the trade-off between data rate and coverage range, LoRaWAN systems can utilize higher SF for longer coverage range. For near-field communication systems, the value of SF is selected as 7 to 9 [28] and [32].

To be explicit, we first group the block-hash H_B into SF bits as a vector $\mathbf{K}$ via the weighted matrix $\mathbf{W} \in \mathbb{Z}^{L_B \times \lceil \frac{L_B}{SF} \rceil}$, which is composed by the repeated weighted vector $\mathbf{wd} = [2^0, 2^1, \dots, 2^{SF-1}]^T$ given as below.

$$\begin{aligned}
 \mathbf{K} &= H_B \mathbf{W}, \quad \mathbf{K} \in [0, 2^{SF} - 1] \\
 \mathbf{W} &= \text{diag}(\mathbf{wd}, \dots, \mathbf{wd})
 \end{aligned} \quad (7)$$

Then the grouped the block-hash $\mathbf{K}$ is modulated by left CS-CSS (LCS) scheme by [28]:

$$\begin{aligned}
 2^{SF} &= B \cdot T_s (SF \in \mathbb{Z} \cap [7, 12]) \\
 f_i(t) &= \begin{cases} fc + \frac{Bw}{T_s} \left(t + \frac{K_i}{Bw} \right), & t \in \left[-\frac{T_s}{2}, \frac{T_s}{2} - \frac{K_i}{Bw} \right] \\ fc + \frac{Bw}{T_s} \left(t + \frac{K_i}{Bw} \right) - Bw, & t \in \left[\frac{T_s}{2} - \frac{K_i}{Bw}, \frac{T_s}{2} \right] \\ 0, & \text{elsewise} \end{cases}
 \end{aligned}$$

$$\frac{d\phi(t)}{dt} = 2\pi \cdot f_i(t); \quad c_{K_i}(t) = \exp(j\phi(t));$$

$$\mathbf{S}[n] = c_{K_i}(t)|_{t=\frac{n}{Bw}};$$

where Bw , T_s and f_c are respectively the expand bandwidth, chirp duration and central frequency. The instantaneous frequency, phase, waveform are respectively denoted as $f_i(t)$, $\phi(t)$, $c_{K_i}(t)$. $\mathbf{S}[n]$ denotes as sampled symbols sequence.

At the legitimate receiver, after perfect equalization and coherent demodulation [33], [34], [35], the estimated confidential data will be demodulated as $\hat{H}_B$. In order to obtain the chaotic key, the legitimate receiver utilize linear-searching methods to obtain the indexes which meet the predefined conditions $\{\hat{x}_0\}$. Subsequently, receivers apply the same hash function $H^{bc}(\cdot)$ used at the transmitter to generate digital signatures $\{\hat{H}_x\}$.

Then after retrieving block-hash replica via distributed agent nodes $\hat{H}_x$, the receiver can extract the most approximated chaotic key $\hat{x}_0$ via the minimum mean square error (MMSE) detection, by comparing the replica $\hat{H}_x$ with $\{\hat{H}_x\}$. Algorithm 2 presents the details of the chaotic key retrieval.

3) *Two-Level Eavesdropping Models*: Nonetheless, due to broadcasting property of wireless channels, as shown in the Fig. 7, eavesdroppers might try to intercept the confidential information. Without loss of generality, we assume that the eavesdroppers would use the same IoT protocols as those used by legitimate transceivers. According to the information they occasionally learn, the eavesdropping might have two levels.

- At the first level, eavesdroppers Eve^a might learn the chaotic key by utilizing the brute-force algorithm to obtain the key as $\hat{x}_{0,E}^a \in \mathbb{S}^a = [0, L_B - 1] \cap \mathbb{Z}$, where $\hat{x}_{0,E}^a$ denotes the chaotic keys that Eve^a speculate and $\mathbb{S}^a$ denotes the key collections.
- At the second level, eavesdroppers Eve^b might have the prior knowledge that the confidential chaotic key is embedded as the bits-flipping scheme as given by (5). They will firstly demodulate the eavesdropped wireless signal, then use the linear-searching method trying to learn the chaotic key as $\hat{x}_{0,E}^b \in \mathbb{S}^b = \{x \mid \hat{\mathbf{H}}_B^b(L_{Mine} + x + 1) = 0, x \in [0, L_B - L_{Mine} - 1] \cap \mathbb{Z}\}$, where $\hat{x}_{0,E}^b$ denotes once-speculate key of Eve^b , $\hat{\mathbf{H}}_B^b$ is the demodulated data.

Notably, the eavesdroppers could not retrieve the registered information in the blockchain. Even though the eavesdroppers might put efforts to conduct the non-coherent detection and brute-force method, thanks to the consensus mechanism design, the confidential chaotic key can not be obtained by legitimate users. Hence the following data transmissions could be protected and the security is guaranteed.

Subsequently, after the Join Procedure finishes, the nodes would deliver the user data in a chaotic scrambling way in the Transmit Procedure.

III. THEORETICAL PERFORMANCE ANALYSIS

In this section, we present the theoretical reliability, security and authentication performances, and then the complexity analysis is provided.

A. Bit Error Rate Analysis

Without loss of generality, we assume the coherent detection is conducted. Then the received signal at the legitimate receiver can be modeled as:

$$\hat{\mathbf{s}}(t) = \sum_{i=1}^{N_{mp}} \alpha_i \mathbf{s}(t - \lambda_i) + \mathbf{n}_i; \mathbf{n}_i \sim \mathcal{CN}(0, 2\sigma^2); \quad (9)$$

where N_{mp} denotes the number of multipaths and $\mathbf{s}(t)$ represents the transmitted signal. Besides, α_i and λ_i respectively represent the fading coefficient and the delay of the i_{th} path, $\mathbf{n}_i$ denotes the complex white Gaussian noise, which follows the circular normal distribution with the zero mean and the variance σ^2 , i.e., $\mathcal{CN}(0, \sigma^2)$. In the special case that $N_{mp} = 1$ and $\alpha_i = 1$, we will obtain the additive white Gaussian noise (AWGN) channel.

For the proposed LoRaWAN system, the bit error rate (BER) of legitimate users applying the coherent detection over AWGN and Rayleigh fading channel are denoted by P_e^{AWGN} , which could be approximated by [35]:

$$P_e^{AWGN} \approx \frac{1}{2} Q \left(\frac{\sqrt{\eta} - \sqrt{[4](\hbar_{2SF-1})^2 - \frac{\pi^2}{12}}}{\sqrt{\hbar_{2SF-1} - \sqrt{(\hbar_{2SF-1})^2 - \frac{\pi^2}{12} + 0.5}}} \right) \quad (10)$$

$$P_e^{Rayleigh} \approx \frac{1}{2} \left[Q(-\sqrt{2\hbar_{2SF-1}}) - \sqrt{\frac{\eta}{\eta+1}} e^{-\frac{\hbar_{2SF-1}}{\eta+1}} \times Q\left(-\sqrt{\frac{2\eta}{\eta+1} \hbar_{2SF-1}}\right) \right] \quad (11)$$

where $\hbar_m = \sum_{k=1}^m \frac{1}{k} \approx \ln(m) + \frac{1}{2m} + 0.57722$ is the Harmonic number and 0.57722 is the Euler-Mascheroni constant. Notably, η denotes the average signal-to-noise rate (SNR).

Similarly, we can derive the BER expressions for eavesdroppers. Assuming that the malicious users also apply the non-coherent detection, then the BER performances over AGWN and Rayleigh fading channels can be lower bounded and approximated, which are both obviously higher than that of legitimate users performing coherent detection in [35].

$$P_{e,E}^{AWGN} = \frac{2^{SF-1}}{2^{SF}-1} \sum_{k=1}^{2^{SF}-1} \left[\frac{(-1)^{k+1}}{k+1} \times \binom{2^{SF}-1}{k} \exp\left(-\frac{k}{k+1}\eta\right) \right]$$

$$P_{e,E}^{Rayleigh} = \frac{2^{SF-1}}{2^{SF}-1} \sum_{k=1}^{2^{SF}-1} \left[(-1)^{k+1} \times \binom{2^{SF}-1}{k} \frac{1}{1+(k+1)\eta} \right] \quad (12)$$

B. Security Performance

Under assumptions of perfect frame synchronization, equalization and coherent demodulation, the security performances are analyzed in terms of the authentication error rate, the mutual information and the secrecy capacity.

1) *Authentication Accuracy Analysis*: For confidential chaotic data $\mathbf{H}_B$, with the bits-flipping scheme, the authentication error rate (AER) of legitimate receivers P_a can be expressed as

$$P_a = 1 - \mathbb{P} \left(\hat{H}_B(\hat{x}_0 + L_{Mine} + 1) = \tilde{H}_B(x_0 + L_{Mine} + 1) \right)$$

$$\approx 1 - \sum_{k=0}^{L_B-1} \binom{L_B-1}{k} P_e^k (1 - P_e)^{L_B-k} = P_e \quad (13)$$

Then, we denote $P_{e,E}$ as the BER of near-field eavesdropping. Besides, the authentication accuracy rate of the above the eavesdropper Eve^a and Eve^b are respectively denoted as P'_{a,E^a} and P'_{a,E^b} , which can be expressed as below.

$$P'_{a,E^a} = \mathbb{P} \left(\hat{x}_{0,E}^a = x_0 \right) = \frac{1}{\text{Card}(\mathcal{S}^a)} = \frac{1}{L_B}$$

$$P'_{a,E^b} = \mathbb{P} \left(x_0 \in \{\hat{x}_{0,E}^b\} \mid \hat{H}_B^b(\{\hat{x}_{0,E}^b\}) = 0 \right) = \frac{1 - P_{e,E}}{\text{Card}(\{\hat{x}_{0,E}^b\})} \quad (14)$$

where $\text{Card}(\cdot)$ is the cardinality function of a set. Therefore, the authentication error rate (AER) of the above two types of eavesdroppers can be calculated as $P_{a,E^a,b} = 1 - P'_{a,E^a,b}$.

From (12)–(14), we can notice that P_{a,E^a} keep invariant for the key space. Moreover, since $Card(\{\hat{x}_{0,E}^b\}) \gg 1$ and $P_e \lesssim P_{e,E}$, the AER of legitimate users is much lower than that of eavesdroppers, i.e., $P_{a,E^a,b} \gg P_a$. Hence, with the proposed blockchain enable authentication design, the security performance can be improved.

2) *Mutual Information and Secrecy Capacity Analysis:* Next we analyze the security performances in terms of the secrecy capacity. Without loss of generality, we assume that confidential chaotic key are uniformly distributed in the chaotic key space S , where $Card(S) = L_B - L_{Mine}$.

Then, for the N strong symmetric channel (N-SSC), we evaluate the mutual information between the transmitted chaotic key x_0 and the recovered one at the legitimate receivers $\hat{x}_0$ as well as eavesdroppers $\hat{x}_{0,E}$ as below.

$$\begin{aligned} I_n(\hat{x}_0; x_0) &= H(x_0) - H(x_0|\hat{x}_0) \\ &= lb(Card(S)) + (1 - P_a)lb(1 - P_a) + P_a lb\left(\frac{P_a}{Card(S) - 1}\right) \\ I_{n,E^a,b}(\hat{x}_{0,E^a,b}; x_0) &= H(x_0) - H(x_0|\hat{x}_{0,E^a,b}) \\ &= lb(Card(S)) + (1 - P_{a,E^a,b})lb(1 - P_{a,E^a,b}) \\ &\quad + P_{a,E^a,b} lb\left(\frac{P_{a,E^a,b}}{Card(S^{a,b}) - 1}\right) \end{aligned} \quad (15)$$

where $I_n(\hat{x}_0; x_0)$ and $I_{n,E^a,b}(\hat{x}_{0,E^a,b}; x_0)$ respectively represent the mutual information of the chaotic key at the legitimate receivers and eavesdroppers, $H(\cdot)$ indicates the entropy operation, and $lb(\cdot)$ denotes the 2-base logarithm.

Additionally, the secrecy capacity $C_{Secrecy}$ of the legitimate users is represented as below [36]:

$$C_{Secrecy} = I_n(\hat{x}_0; x_0) - I_{n,E^a,b}(\hat{x}_{0,E^a,b}; x_0) \quad (16)$$

Notably, thanks to the anti-tamper property of the Hash function, the secret chaotic keys can be protected. Although malicious users might apply the time-consuming brute force methods trying to retrieve the user data, since the chaotic sequence has the property of being aperiodic, the user data can hardly be retrieved in real time, hence the security performances can be improved.

C. Throughput and Delay

Subsequently, we derive the delay and the throughput per second (TPS) by applying the linear-fitting methods as below.

Let N^{tl} denote the total number of terminals to access the network, τ_Σ denote the total delay at the transceiver in Join Procedure, while the average delay of each IoT device is denoted as $\hat{\tau}$. Since the delay is linearly dependent on the key-space $L_s = Card(S) = L_B - L_{Mine}$ with the growth parameter τ_b , then we have:

$$\begin{aligned} \tau_\Sigma &\approx N^{tl} \cdot \tau + \tau_0 + \varepsilon, \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2) \\ \hat{\tau} &\approx \frac{\sum_{i=1}^T \tau_{\Sigma_i} (N_i^{tl} - \overline{N^{tl}})}{\sum_{i=1}^T (N_i^{tl} - \overline{N^{tl}})^2}, \hat{\tau}_b = \frac{\hat{\tau}}{L_s} \end{aligned} \quad (17)$$

TABLE I
SIMULATION PARAMETER SETTINGS

Classes	Type	Setting
Blockchain	H^{bc}	MD2 ($L_B=128$, Hex)
	H^s	MD2 ($L_B=128$, Ascii)
	$MineStr$	bc(Hex)
	$MineStart$	-352.213(Randomly)
Transmission	$MineStep$	0.326(Randomly)
	CS-CSS	SF=8; $B_w=125$ kHz
	N^{tl}	200,000
	AWGN Channel	$\eta_{CSS} = -15$ dB

where τ_0 denotes the delay induced by the signal processing at transceivers, ε is the estimation error vector in the linear-fitting which follows the normal distribution. Besides, T is the number of replicate tests, N_i^{tl} denotes the number of terminals at the i_{th} test, and $\overline{N^{tl}} = \frac{1}{T} \sum_{i=1}^T N_i^{tl}$ represents the average number of terminals.

Moreover, we define the throughput per second (TPS) as the average accomplished transmission messages, which is the reciprocal of the delay. Then we derive the average TPS in terms of the each framing rate $\hat{R}_f$ and the bit rate as $\hat{R}_b$:

$$\begin{aligned} \hat{R}_f &= \frac{1}{\hat{\tau}} = \frac{\sum_{i=1}^T (N_i^{tl} - \overline{N^{tl}})^2}{\sum_{i=1}^T \tau_{\Sigma_i} (N_i^{tl} - \overline{N^{tl}})} \\ \hat{R}_b &= L_s \hat{R}_f = \frac{L_s}{\hat{\tau}} = \frac{1}{\hat{\tau}_b} \end{aligned} \quad (18)$$

IV. SIMULATIONS AND IMPLEMENTATIONS

In this section, we present the simulation results of our proposed chaotic scrambling IoT communications with the blockchain-based key management and authentication mechanism. In the simulations, the reliability, security and throughput performances are evaluated with Intel(R) Core(TM) i7-8700 CPU@3.200 GHz. Then we implement the proposed design with the STM32F4 board mounted with the ATK-LoRa-01 module and the SX1278 chip. Table I presents the simulation parameter settings.

A. Security Performances

We first investigate the security performances when the spread factor (SF) has different values.

As shown in Fig. 8, the theoretical AER performances match simulated results when $SF \in [7, 8, 9, 10, 11, 12]$. Besides, we can notice that when SF increases larger spreading gain can be attained and better security performances can be achieved. Moreover, we provide the AER performances when two different types of near-field attacks are imposed. It can be observed from the figure that the AER of Eve^a is higher than that of Eve^b , while the AER of both eavesdroppers is higher than that of legitimate users.

Furthermore, we compare the performance of the proposed design with the benchmark scheme following the standard IoT protocol, which deliver the confidential chaotic key via the dedicated channel with the direct transmission (DT). From the

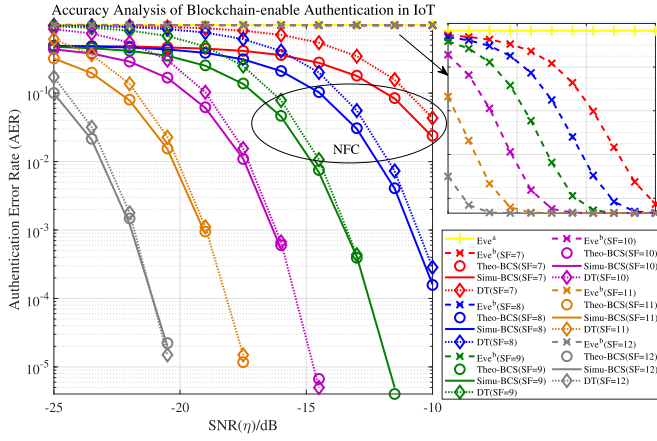


Fig. 8. Authentication accuracy performances of legitimate users and eavesdroppers over AWGN channel with different $SF \in [7, 8, 9, 10, 11, 12]$, and performances under NFC with $SF = 7, 8, 9$.

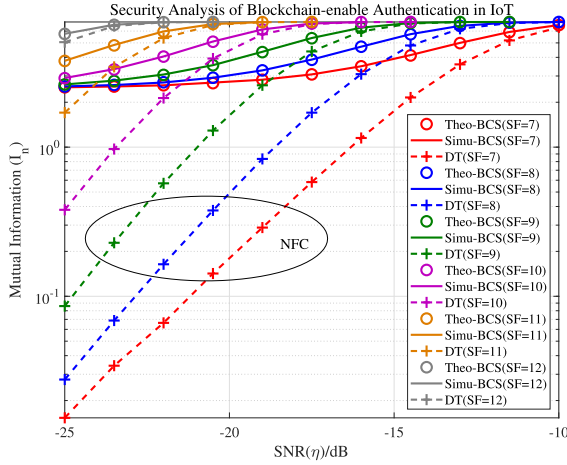


Fig. 9. Mutual information of legitimate users and eavesdroppers over AWGN channel with different $SF \in [7, 8, 9, 10, 11, 12]$, and performances under NFC with $SF = 7, 8, 9$.

figure, we can observe that the proposed design can achieve higher security performances with lower AER than the DT scheme although the latter one can achieve lower AER than eavesdroppers. Thus with the aid of the proposed chaotic scrambling and the blockchain-based authentication, the eavesdroppers could hardly retrieve the information due to high AER. The reason is that the identity of eavesdroppers can not be authenticated and no knowledge in the blocks can be obtained for the key recovery.

Then in Fig. 9, we compare the mutual information of legitimate users and eavesdroppers over AWGN channel with different $SF \in [7, 8, 9, 10, 11, 12]$. We can notice that by applying the proposed design, the mutual information of the proposed scheme is far higher than the benchmark DT scheme especially when SF has lower value of 7-9, thus the desirable security improvement can be attained to combat the near-field attack.

Moreover, Fig. 10 illustrates the secrecy capacity of the proposed system and benchmark scheme undergoing two types of malicious eavesdropping over AWGN channel when $SF \in [7, 8, 9, 10, 11, 12]$. Similar to the LPWAN communication system in the near-field attack scenario with $SF = 7, 8, 9$ at

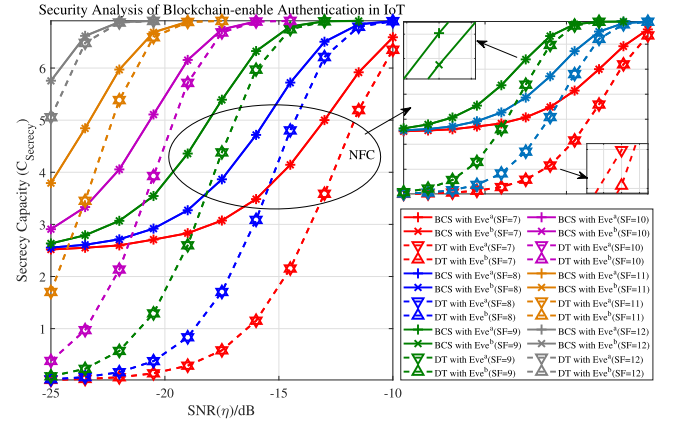


Fig. 10. Secrecy capacity performances of the proposed system undergoing eavesdropping over AWGN channel with different $SF \in [7, 8, 9, 10, 11, 12]$, and performances under NFC with $SF = 7, 8, 9$.

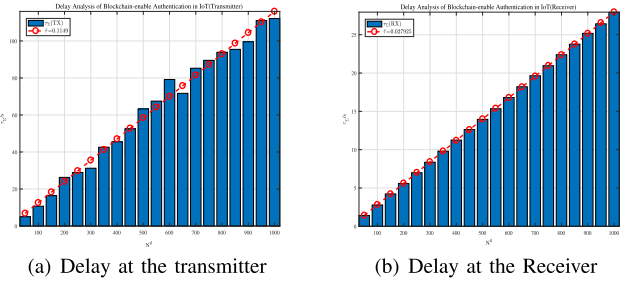


Fig. 11. Delay performances of the proposed system.

lower SNR, we can observe that when the SF has larger value, the secrecy capacity also becomes higher thanks to the enhanced randomness. Besides, we can also notice that the proposed BCS-IoT scheme has much higher secrecy capacity than the benchmark scheme. Thus better security performances can be achieved.

B. Throughput and Delay Performance

The Fig. 11 compares the delay performances induced by the signal processing versus the number of user equipments N^{tl} . In accordance with (17), we can observe from Fig. 11(a) and Fig. 11(b) that the delay is approximate linear to N^{tl} . Besides, at the transmitter, due to the authentication and key management via the blockchain, larger processing delay is required. Then we investigate the throughput performances. As illustrated in Fig. 12(a) and Fig. 12(b), since the transmitter needs to conduct the blockchain computations for the authentication before delivering signals over wireless channels, the throughput of the transmitter is smaller than that of the receiver. Besides, we can notice that the legitimate receiver can retrieve the data bits with a relatively steady throughput.

C. Computational Complexity Analysis

Subsequently, we analyze the computational complexity of the proposed design using different parameters.

Table II compares the complexity, the throughput and the delay of the transmitter when applying different block

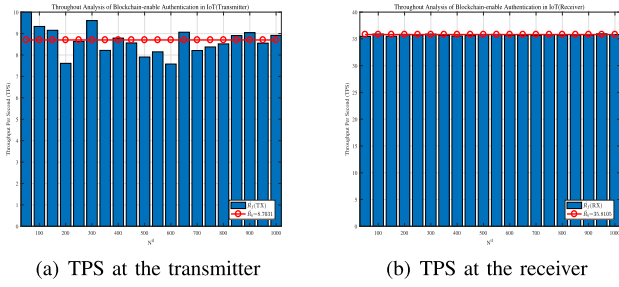


Fig. 12. Throughput performances of the proposed transceiver.

 TABLE II
COMPLEXITY, TPS AND DELAY PERFORMANCE AT THE TRANSMITTER

Condition	MineStr=0(hex)	MineStr=f9(hex)	MineStr=cea(hex)
L_s	124	120	116
$\bar{\gamma}$	0.1747	2.945	47.14
$\hat{\tau}/(s)$	0.0081	0.1119	1.7223
$\hat{R}_b/(\text{bps})$	$4.7238(10^4)$	$3.4388(10^3)$	$2.2306(10^2)$

generation hexadecimal conditions $MineStr = 0, f9, cea$. It can be seen that when the length of $MineStr$, i.e., L_{Mine} increases, due to the increasing complexity of creating new blocks, the transmission data rate will decrease accordingly.

Then we analyze the complexity of the proposed system when different Hash functions are used. From Table III, we can notice that when the function SHA-384 is applied, the complexity is relatively lower while the delay is smaller. Thus L_B and L_s increase and higher data rate can be achieved.

D. Implementations

Finally, we provide the hardware implementation on the STM32F4 board mounted for the proposed chaotic scrambling IoT system with the LoRa communication chip involving the ATK-LoRa-01 module and the SX1278 chip. And as for the module setting, we transmit the signal at 20 dBm with 19.2kbps. Moreover, we set the legitimate transmitter (right bottom), legitimate receiver (right top) and the eavesdropper (left) to have the same module address and channel, which can be regarded as the near-field broadcasting scenario.

From Fig. 13, we can observe that when the legitimate user transmits data “22” with chaotic scrambling index 35, the illegitimate receiver can not obtain the correct data but outputs “8”. By contrast, the legitimate receiver can descramble and conduct the information recovery reliably.

V. CONCLUSION

In this paper, we propose a chaotic scrambling IoT communication system with the blockchain-based key management and authentication. By exploiting the properties of chaotic sequences being aperiodic and sensitive to initial values, and the traceable and tamper-proof property of the blockchain, the user data can be hidden behind irregular chaotic waveforms with the secret chaotic key embedded into anti-tampering blockchain. At the legitimate receiver, after the identity of the IoT device is authenticated via a dual authentication mechanism, the chaotic key can

 TABLE III
COMPLEXITY, TPS AND DELAY PERFORMANCE AT THE TRANSMITTER

Hash	MD2	MD5	SHA1	SHA-256	SHA-384
L_B	128	128	152	248	376
L_s	120	120	144	240	368
$\bar{\gamma}$	2.843	2.863	2.148	1.204	0.760
$\bar{\tau}_b/(\text{ms})$	84.8	80.8	62.0	36.0	23.3
$\hat{R}_b/(\text{bps})$	1178	1238	1613	2778	4288

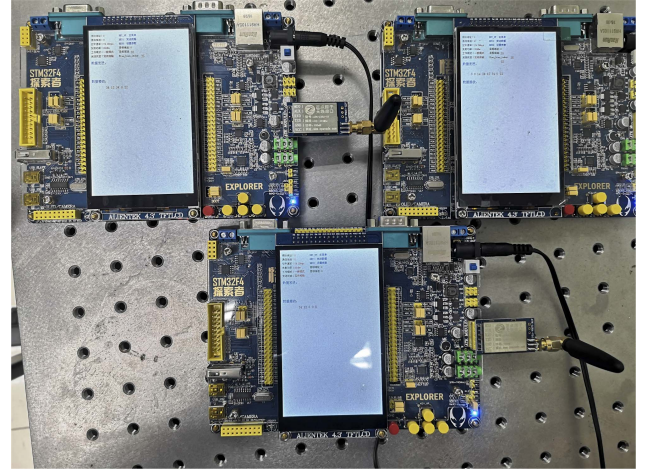


Fig. 13. Practical implementations of the chaotic scrambling IoT communications: Transmitter(right bottom), Legitimate Receiver(right top) and Illegitimate Receiver(left).

be extracted. Then the user data could be recovered reliably. Subsequently, we provide theoretical performance analysis. Simulation results demonstrate that the theoretical Authentication error rate results match the simulated ones. Besides, when the spread factor has larger value, the authentication accuracy rate and the secrecy capacity can be improved, thus the security performances can be enhanced. Moreover, we provide practical implementations of the proposed chaotic scrambling IoT system to validate our design. Therefore, with the aid of the blockchain and chaotic scrambling, the security of IoT can be enhanced to meet the increasing demands of user equipments.

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